

PortPy Planning & Optimization for Radiation Therapy in Python

An Open-Source Platform for Benchmarking & Clinical Translation of Radiotherapy Treatment-Planning Optimization

- **Masoud Zarepisheh**, Gourav Jhanwar, Mojtaba Tefagh, Linda Hong, Qijie Huang, Ying Zhou, Hai Pham, Jie Yang, Seppo Tuomaala, Anthony Magliari, Michael Folkerts

Department of Medical Physics · Memorial Sloan Kettering Cancer Center · New York

Varian Medical Systems, Siemens Healthineers



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Project Goal

Goal: A Python-based, open-source project dedicated to radiotherapy **treatment-planning optimization**.



Benchmark Dataset

~200 lung & 130 prostate cases: CT, structures, IMRT/VMAT plans, and dose-influence matrices.



Baseline Algorithms

IMRT (fluence map, leaf sequencing) and VMAT (direct aperture) — AI and non-AI optimization.



Native Eclipse Integration

Produces Eclipse-compatible optimal fluence; uses Eclipse's beamlet dose calculation.

Three core features — each detailed in the next slides.



Benchmark Dataset



330 curated cases (CT, structures)

200 lung · 130 prostate

DICOM

TCIA



IMRT & VMAT plans

Reference ECHO plans

DICOM



Clinical criteria

e.g., lung mean-dose limits

JSON

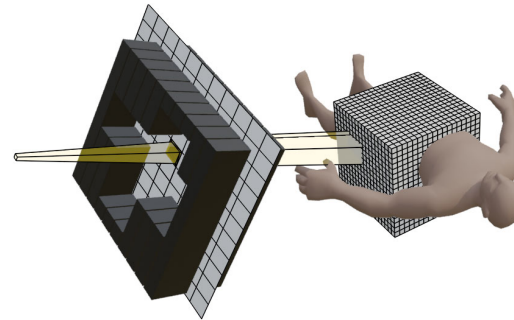


Dose-influence matrix (72 beams)

Beamlet → voxel dose

JSON

HDF5



Pre-computed dose-influence matrix

- Beamlet-to-voxel dose for benchmark dataset



Want to use your own dataset?

Two ways:



Eclipse ESAPI script



AI-based beamlet dose calculation



Baseline Algorithms

1

IMRT fluence-map optimization

output → Eclipse-compatible optimal fluence

2

IMRT fluence-map => Leaf sequencing

output → DICOM-RT

3

VMAT aperture optimization

output → DICOM-RT

4

AI-based dose-prediction planning (IMRT & VMAT)

AI predicted dose => IMRT/VMAT optimization => deliverable plan

output → DICOM-RT

5

Global optimization via Mixed Integer Programming

Gold standard — but computationally expensive (~1 week)

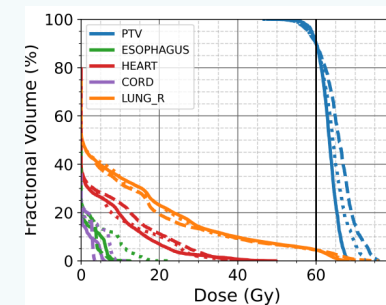
All provided as ready-to-run [Jupyter notebooks](#).



Global optimization — the gold standard



Non-convex problem — heuristics can stall in local optima.



Plan quality (DVH) vs. global optimum (~1 week to solve)

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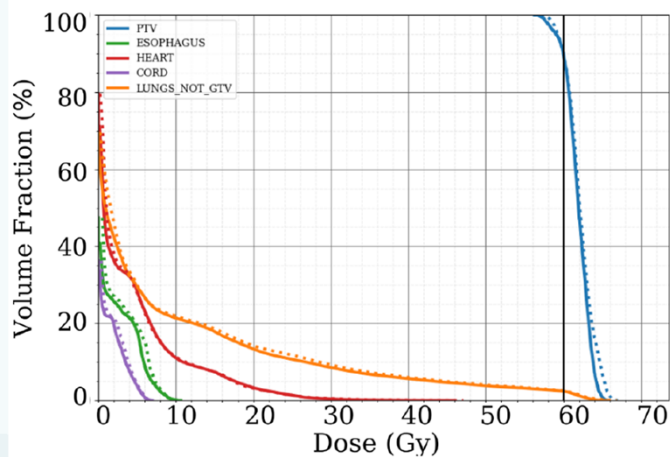
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Eclipse Integration — Clinical Deployment

PortPy plans are **dosimetrically accurate in Eclipse** — and **clinically deployed at MSK**.

✓ Dosimetric accuracy



Imported into Eclipse — **the Eclipse-recomputed dose matches the PortPy plan** with negligible discrepancy.

✓ Clinical deployment

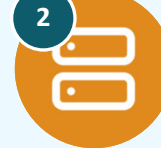


Inside Eclipse



1
Template plan
& run ECHO

MSK Server



2
ECHO optimization



Inside Eclipse



3
Final dose
& plan review

Constrained hierarchical solve · ~30 min

12,000+ patients treated with ECHO at MSK

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Coming Soon — GPU-Accelerated Optimization



Plug in any solver

PortPy works with a wide range of optimization engines out of the box:

OPEN-SOURCE

ECOS

OSQP

SCS

COMMERCIAL

MOSEK

GUROBI

CPLEX

Same code, any backend — switch solvers by changing a single argument.

COMING SOON



**A GPU-accelerated engine,
built for radiotherapy**

~5x

faster than the best commercial solver

Purpose-built for the structure of radiotherapy optimization problems.

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Summary



Powerful, open-source research tool

Fully customizable IMRT/VMAT optimization engine for radiotherapy.



Benchmark dataset

~200 lung and 130 prostate curated cases.



Baseline algorithms

IMRT (fluence map, leaf sequencing), VMAT (aperture), and global optimization.



Eclipse integration — clinically deployed

12,000+ patients treated with ECHO at MSK.



GPU-accelerated optimization coming soon

~5× faster than the best commercial solver.



40,000+ downloads (~1,000/month)



github.com/PortPy-Project



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